

BEC303 Network Analysis — Assignment 1 —

Systematic analysis of DC and AC networks

Module 1: Basic Concepts — Mesh and Node Analysis

Field	Value
Maximum marks	20
Weightage	10% of CIE
Issue date	<<TBD>>
Due date	<<TBD>>
Items	10 problems + 3 theory questions
CO / Bloom coverage	CO1: 13 · L1–L2 0, L3–L4 13, L5–L6 0

Objective

Build fluency in choosing and executing the correct systematic analysis method — nodal or mesh, with supernode and supermesh repairs — and in carrying that method unchanged into the phasor domain.

Learning outcomes

- Student will be able to select nodal or mesh analysis on the basis of node and mesh counts and source types.
- Student will be able to construct supernode and supermesh equations and justify why the equation count is preserved.
- Student will be able to convert a sinusoidal network to phasor form and solve it with the same systematic method.
- Student will be able to apply source transformation and delta-wye conversion as preprocessing steps and state their limits.

Part A — Problems (17 marks)

P1. For a three-node resistive network of your own construction containing one grounded voltage source and one current source, write the full nodal equation set and solve for all node voltages. State the number of equations before you write any of them and justify the count. [C01 | L3 | 2 marks]

P2. Modify the P1 network so that the voltage source now floats between two non-reference nodes. Solve again using a supernode, and verify that the node voltages differ from P1 in the way you predict. [C01 | L3 | 2 marks]

P3. Solve the P1 network a second time by mesh analysis. Compare the number of equations required by each method and state which method you would choose for this topology and why. [C01 | L4 | 2 marks]

P4. Construct a two-mesh circuit in which a current source lies on the shared branch. Form the supermesh, solve, and then recover the voltage across the current source from the original mesh equations. [C01 | L3

| 2 marks]

P5. Given a network containing a VCCS whose control voltage is not a node voltage, write the nodal equations, the auxiliary equation, and the final substituted system. Point out explicitly where matrix symmetry is lost. [C01 | L4 | 2 marks]

P6. Reduce a network containing two voltage sources with series resistances to a single loop using repeated source transformations. Then compute the power dissipated inside the original sources and explain why this cannot be read off the transformed circuit. [C01 | L4 | 2 marks]

P7. Take a Wheatstone bridge with five unequal resistors and a supply. Show that series-parallel reduction stalls, apply one delta-wye conversion, and compute the supply current. [C01 | L3 | 2 marks]

P8. Convert $v_1 = 8\sin(500t + 15^\circ)$ and $v_2 = 6\cos(500t - 40^\circ)$ to phasors, add them, and write the sum back in the time domain. Show the intermediate rectangular form. [C01 | L3 | 1 marks]

P9. For a series RLC branch with $R = 50\ \Omega$, $L = 20\ \text{mH}$ and $C = 10\ \mu\text{F}$, tabulate the impedance at $\omega = 500$, 2236 and 5000 rad/s, and comment on which element dominates at each frequency. [C01 | L4 | 1 marks]

P10. Solve a two-node AC network containing one dependent source by phasor nodal analysis, and present the answer as both a phasor and a time-domain waveform. [C01 | L4 | 1 marks]

Part B — Theory (3 marks)

T1. Explain why the Nth KCL equation in an N-node circuit carries no new information, and demonstrate this on a three-node example by showing that one equation is the negated sum of the others. [C01 | L4 | 1 marks]

T2. Argue in no more than half a page why phasor analysis is exact rather than approximate, and state precisely what part of the circuit response it cannot describe. [C01 | L4 | 1 marks]

T3. Compare the supernode and the supermesh: what breaks in each case, what repair is applied, and why the total equation count is unchanged in both. [C01 | L4 | 1 marks]

Submission guidelines

- Format: PDF, single file
- File name: <USN>_BEC303_Assignment1.pdf
- Channel: LMS upload
- Late policy: 10% per day, nothing after 5 days
- Every numerical answer must carry units and a stated reference direction.

Assessment rubric

Criterion	Weight	Excellent	Good	Satisfactory	Needs improvement
Correctness of equations and solution	8	All equations correctly formed with consistent sign conventions; every numerical answer correct and checked.	Equations correct; at most one arithmetic slip that does not change the conclusion.	Most equations correct but sign conventions inconsistent, or several numerical errors.	Equations incorrectly formed; branch currents written without difference terms.
Method selection and justification	5	Method chosen with an explicit count of equations and a stated reason; alternative considered and rejected on evidence.	Method chosen with a reason given, alternative mentioned.	Method used without justification.	Method appears arbitrary or switches mid-problem without explanation.
Handling of special cases (supernode, supermesh, dependent sources)	4	Special cases identified immediately, repair applied correctly, and the equation-count argument stated.	Special cases handled correctly with minor gaps in justification.	Special case recognised but repair partially incorrect.	Special case not recognised; unknown source current or voltage left in the equations.
Presentation, units and phasor conventions	3	Every quantity carries units and a reference direction; one phase convention used throughout and stated.	Units present throughout, conventions consistent, minor labelling gaps.	Units intermittent or conventions stated only implicitly.	Units missing; sine and cosine forms mixed without conversion.
Total	20				

Course outcome attainment

Course outcome	Marks carried
CO1	20
Total	20

Academic integrity

Individual submission. Identical submissions score zero for both parties.